Chapter 12 Further Calculus - Homework Answers

Chapter revision | Warm-up

1 a $3\cos 3x$ b $2x\cos x - x^2\sin x$ c $2x\sec^2(x^2)$

2 a $\frac{1}{2}\sin 2x + C$ b $\left[\tan x\right]_0^{\frac{\pi}{4}} = 1$

3 a $\frac{\pi}{3}$ b $\frac{2\pi}{3}$ c $-\frac{\pi}{4}$

4 $y = \sin^{-1} x$: from $\left(-1, -\frac{\pi}{2}\right)$ to $\left(1, \frac{\pi}{2}\right)$ through the origin. $y = \cos^{-1} x$: from $(-1, \pi)$ to $(1, 0)$ through $\left(0, \frac{\pi}{2}\right)$. $y = \tan^{-1} x$: through the origin, with horizontal asymptotes $y = \pm\frac{\pi}{2}$. Domains and ranges as in the summary table.

5 $\frac{1}{4}(x^2+1)^4 + C$

6 $\left[\frac{1}{3}x^3\right]_0^3 = 9$ square units

Chapter 12 Further Calculus - Homework Answers

Lesson 12A | Inverse trigonometric functions: differentiating

Lesson Examples

C1 **Tutor demo.** a $\sin^{-1}x + \dfrac{x}{\sqrt{1-x^2}}$ b $-\dfrac{3}{\sqrt{1-(3x-1)^2}} = -\dfrac{3}{\sqrt{6x-9x^2}}$

Your turn. a $2x\tan^{-1}x + \dfrac{x^2}{1+x^2}$ b $\dfrac{2}{1+(2x+1)^2} = \dfrac{1}{2x^2+2x+1}$

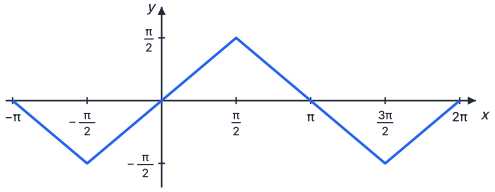
C2 **Tutor demo.** $\dfrac{1}{\sqrt{1-(x+1)^2}} = 2$ gives $(x+1)^2 = \frac{3}{4}$, so the points are $\left(-1+\frac{\sqrt{3}}{2}, \frac{\pi}{3}\right)$ and $\left(-1-\frac{\sqrt{3}}{2}, -\frac{\pi}{3}\right)$.

Your turn. At $x=1$: $y=\frac{\pi}{4}$, $y'=\frac{1}{2}$. Tangent: $y=\frac{1}{2}x-\frac{1}{2}+\frac{\pi}{4}$.

C3 **Tutor demo.** With $u=\dfrac{x-1}{x+1}$, $u'=\dfrac{2}{(x+1)^2}$ and $1+u^2=\dfrac{2(x^2+1)}{(x+1)^2}$, so $\dfrac{d}{dx}\tan^{-1}u=\dfrac{1}{1+x^2}$ and $f'(x)=0$. Then $f(x)=f(0)=0-\tan^{-1}(-1)=\frac{\pi}{4}$.

Your turn. $\dfrac{d}{dx}\tan^{-1}\frac{1}{x}=\dfrac{-1/x^2}{1+1/x^2}=-\dfrac{1}{1+x^2}$, so the sum has zero derivative. For $x>0$ (put $x=1$): $\frac{\pi}{2}$. For $x<0$ (put $x=-1$): $-\frac{\pi}{2}$.

C4 **Tutor demo.** $y'=\dfrac{\cos x}{\sqrt{1-\sin^2 x}}=\dfrac{\cos x}{|\cos x|}$, so $y'=1$ where $\cos x>0$ and $y'=-1$ where $\cos x<0$. The graph zigzags between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, through $(-\pi,0)$, $\left(-\frac{\pi}{2},-\frac{\pi}{2}\right)$, $\left(\frac{\pi}{2},\frac{\pi}{2}\right)$, $\left(\frac{3\pi}{2},-\frac{\pi}{2}\right)$, $(2\pi,0)$.



C5 **Tutor demo.** a $\dfrac{2e^{2x}}{1+e^{4x}}$ b $\dfrac{2x}{\sqrt{1-x^4}}$ **Your turn.** a $-\dfrac{3x^2}{\sqrt{1-x^6}}$ b $\dfrac{2x}{1+(x^2+1)^2}$

Textbook Homework

1 a $\dfrac{-1}{\sqrt{1-x^2}}$ b $\dfrac{2}{\sqrt{1-4x^2}}$ c $\dfrac{-5}{\sqrt{1-25x^2}}$ d $\dfrac{2x}{\sqrt{1-x^4}}$ e $\dfrac{1}{x^2+4x+5}$
f $\sin^{-1}x+\dfrac{x}{\sqrt{1-x^2}}$ g $2x\tan^{-1}x+1$ h $\dfrac{-1}{2\sqrt{x-x^2}}$ i $\dfrac{-1}{1+x^2}$

2 a 2 b 2 c 1 d -1

3 a Tangent $y=-6x+\pi$, normal $y=\frac{1}{6}x+\pi$. b Tangent $y=\frac{1}{\sqrt{2}}x+\frac{\pi}{4}-1$, normal $y=-\sqrt{2}x+\frac{\pi}{4}+2$.

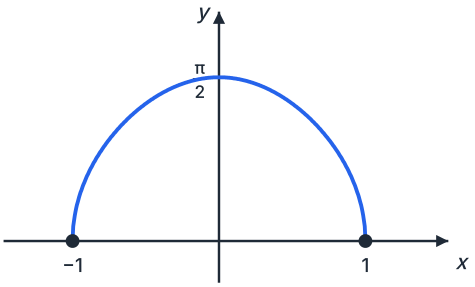
4 a π b $\frac{\pi}{2}$

5 b concave up

6 a $\cos^{-1}x$ b $\dfrac{3e^{3x}}{\sqrt{1-e^{6x}}}$ c $\dfrac{2}{\sqrt{7+12x-4x^2}}$ d $\dfrac{1}{2\sqrt{1-x^2}\sin^{-1}x}$ e $\dfrac{\sin^{-1}\sqrt{1-x}}{2\sqrt{x}}-\dfrac{1}{2\sqrt{1-x}}$

7 a $-\frac{\pi}{2}\le y\le\frac{\pi}{2}$ b $x=\frac{1}{2}\sin y$ c $\dfrac{dx}{dy}=\frac{1}{2}\cos y\ge 0$ for $-\frac{\pi}{2}\le y\le\frac{\pi}{2}$ d $\dfrac{dy}{dx}=\dfrac{2}{\sqrt{1-4x^2}}$

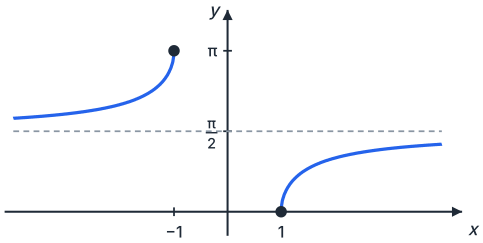
8 **a** $-1 \leq x \leq 1$, even **b** the y-axis, because the function is even **c** $\dfrac{-2x}{\sqrt{1-x^4}}$ **e** the tangents at $x = 1$ and $x = -1$ are vertical **f**



9 **a** $\theta = \angle TEP - \angle BEP$, with $\tan \angle TEP = \frac{4}{x}$ and $\tan \angle BEP = \frac{3}{x}$. **b** $\dfrac{d\theta}{dx} = -\dfrac{4}{x^2+16} + \dfrac{3}{x^2+9} = 0$ gives $x^2 = 12$. **c** $\tan\theta = \dfrac{\frac{4}{x} - \frac{3}{x}}{1 + \frac{12}{x^2}} = \dfrac{x}{x^2+12} = \dfrac{2\sqrt{3}}{24} = \dfrac{\sqrt{3}}{12}$.

10 **c** $\frac{1}{45}$ rad/s

11 **a** $x \geq 1$ or $x \leq -1$ **c** They are undefined. **d** When $x > 1$, $f'(x) = \dfrac{1}{x\sqrt{x^2-1}}$; when $x < -1$, $f'(x) = \dfrac{-1}{x\sqrt{x^2-1}}$. **e** $f'(x) > 0$ for $x > 1$ and for $x < -1$. **f** (i) $\frac{\pi}{2}$ (ii) $\frac{\pi}{2}$ **g**



12 **a** domain: all real x; range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$; odd **c** No, because $\frac{0}{0}$ is undefined. **d** $f'(x) = 1$ when $\cos x > 0$, and $f'(x) = -1$ when $\cos x < 0$. **e** the sawtooth graph shown for the Concept 4 tutor demo, continued in both directions.

Chapter 12 Further Calculus - Homework Answers

Lesson 12A | Inverse trigonometric functions: differentiating

Past-Paper Homework - Worked Solutions

PP 1 North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q1

$\dfrac{d}{dx}\tan^{-1}(2x) = \dfrac{2}{1+(2x)^2} = \dfrac{2}{1+4x^2}$. **Answer D.** C forgets the factor 2 from the chain rule; A and B confuse $\tan^{-1}$ with a power or with $\tan$.

PP 2 The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q4

$$y' = 2x\sin^{-1}(2x) + x^2 \times \frac{2}{\sqrt{1-4x^2}} = 2x\sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$$

Answer (A). (B) drops the factor 2; (C) and (D) square $2x$ incorrectly as $2x^2$.

PP 3 Normanhurst Boys' High School, 2021, Mathematics Extension 1 (Year 12 Course) Assessment Task 4 (Trial Examination), Q5

$$y' = -\left(-\frac{\frac{1}{x}}{\sqrt{1-(\ln x)^2}}\right) = \frac{1}{x\sqrt{(1+\ln x)(1-\ln x)}}$$

Answer (D). The minus signs cancel, and $1-(\ln x)^2$ factorises as a difference of squares (it is not $1-2\ln x$).

PP 4 Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q4

At $x=0$, $y = \cos^{-1}0 = \frac{\pi}{2}$ and $y' = -\dfrac{1}{\sqrt{1-0}} = -1$. Tangent: $y - \frac{\pi}{2} = -x$, that is, $2x+2y-\pi=0$. **Answer B.**

PP 5 Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(a)

$$\frac{d}{dx}\tan^{-1}(\log_e x) = \frac{\frac{1}{x}}{1+(\log_e x)^2} = \frac{1}{x\left(1+(\log_e x)^2\right)}$$

PP 6 Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(a)(i)

$$y' = 3\sin^{-1}(2x) + 3x \times \frac{2}{\sqrt{1-4x^2}} = 3\sin^{-1}(2x) + \frac{6x}{\sqrt{1-4x^2}}$$

PP 7 Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)

$$y' = \cos^{-1}4x + x \times \left(-\frac{4}{\sqrt{1-16x^2}}\right) = \cos^{-1}4x - \frac{4x}{\sqrt{1-16x^2}}$$

PP 8 Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)

$$\frac{d}{dx}\log_e\left(\cos^{-1}x\right) = \frac{\frac{d}{dx}\cos^{-1}x}{\cos^{-1}x} = -\frac{1}{\sqrt{1-x^2}\,\cos^{-1}x}$$

PP 9 Killara High School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q12(d)

$y' = -\dfrac{1}{\pi} \times \dfrac{2}{\sqrt{1-4x^2}}$. At $x=\frac{1}{4}$: $\sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}$, so $m = -\dfrac{2}{\pi}\times\dfrac{2}{\sqrt{3}} = -\dfrac{4}{\sqrt{3}\,\pi}$.

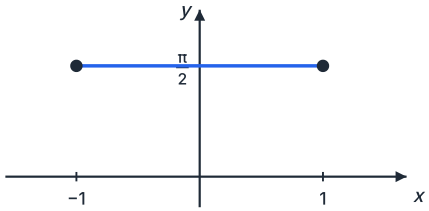
(Check: P is on the curve, since $\frac{1}{2} - \frac{1}{\pi}\sin^{-1}\frac{1}{2} = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$.)

$$y - \tfrac{1}{3} = -\frac{4}{\sqrt{3}\,\pi}\left(x - \tfrac{1}{4}\right) \quad\Longrightarrow\quad y = -\frac{4}{\sqrt{3}\,\pi}x + \frac{1}{\sqrt{3}\,\pi} + \frac{1}{3}$$

PP 10 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q13(a)**

(i) $f'(x) = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{1}{\sqrt{1-x^2}} = 0$ for $-1 < x < 1$, so $f(x)$ is constant. $f(0) = 0 + \frac{\pi}{2}$, so $f(x) = \frac{\pi}{2}$ (this also holds at $x = \pm 1$: $\frac{\pi}{2} + 0$ and $-\frac{\pi}{2} + \pi$).

(ii) A horizontal segment $y = \frac{\pi}{2}$ for $-1 \leq x \leq 1$, with closed endpoints.



Chapter 12 Further Calculus - Homework Answers

Lesson 12B | Inverse trigonometric functions: integrating

Lesson Examples

C1 **Tutor demo.** a $\left[\sin^{-1}x\right]_0^{\frac{\sqrt{3}}{2}} = \frac{\pi}{3}$, or $\left[-\cos^{-1}x\right]_0^{\frac{\sqrt{3}}{2}} = -\frac{\pi}{6}+\frac{\pi}{2} = \frac{\pi}{3}$ b $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$
Your turn. a $\frac{\pi}{4}$ b $\frac{\pi}{4}-\left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$

C2 **Tutor demo.** a $3\sin^{-1}\frac{x}{4}+C$ b $\frac{1}{5\times 2}\tan^{-1}\frac{2x}{5}+C = \frac{1}{10}\tan^{-1}\frac{2x}{5}+C$
Your turn. a $2\sin^{-1}\frac{x}{5}+C$ b $\frac{1}{6}\tan^{-1}\frac{2x}{3}+C$

C3 **Tutor demo.** a $f(x)=e^{2x}$, $f'(x)=2e^{2x}$: $\frac{1}{2}\tan^{-1}(e^{2x})+C$. b $f(x)=\sin x$: $\sin^{-1}(\sin x)+C$, which equals $x+C$ on $-\frac{\pi}{2}<x<\frac{\pi}{2}$ (there the integrand is $\frac{\cos x}{|\cos x|}=1$).
Your turn. $f(x)=x^2$: $\frac{1}{2}\tan^{-1}(x^2)+C$

C4 **Tutor demo.** a $\tan^{-1}x+\dfrac{x}{1+x^2}$. b $\int\tan^{-1}x\,dx = x\tan^{-1}x-\frac{1}{2}\ln(1+x^2)+C$, so $\displaystyle\int_0^1\tan^{-1}x\,dx = \frac{\pi}{4}-\frac{1}{2}\ln 2$.
Your turn. a $\cos^{-1}x-\dfrac{x}{\sqrt{1-x^2}}+\dfrac{x}{\sqrt{1-x^2}} = \cos^{-1}x$. b $\left[x\cos^{-1}x-\sqrt{1-x^2}\right]_0^1 = 0-(-1) = 1$.

Textbook Homework

1 a $\cos^{-1}x+C$ b $\sin^{-1}\frac{x}{2}+C$ c $\frac{1}{3}\tan^{-1}\frac{x}{3}+C$ d $\sin^{-1}\frac{3x}{2}+C$ e $\frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+C$ f $\cos^{-1}\frac{x}{\sqrt{5}}+C$

2 a $\frac{\pi}{2}$ b $\frac{\pi}{8}$ c $\frac{\pi}{4}$ d $\frac{\pi}{12}$ e $\frac{\pi}{6}$ f $\frac{5\pi}{12}$

3 a $y=\sin^{-1}x+\pi$ b $y=\tan^{-1}\frac{x}{4}+\frac{\pi}{4}$

4 a $\frac{\pi}{3}$ b $\frac{\pi}{4}$

5 a $\frac{1}{2}\sin^{-1}2x+C$ b $\frac{1}{4}\tan^{-1}4x+C$ c $\frac{1}{\sqrt{2}}\cos^{-1}\sqrt{2}x+C$ d $\frac{1}{3}\sin^{-1}\frac{3x}{2}+C$ e $\frac{1}{15}\tan^{-1}\frac{3x}{5}+C$ f $\frac{1}{2}\cos^{-1}\frac{2x}{\sqrt{3}}+C$

6 a $\frac{\pi}{18}$ b $\frac{\pi}{12}$ c $\frac{2\pi}{9}\sqrt{3}$ d $\frac{5\pi}{24}$

7 c $\left(\frac{\pi}{12}+\frac{1}{2}\sqrt{3}-1\right)$ unit²

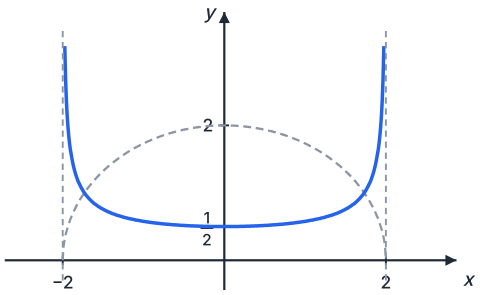
8 b $\left[\cos^{-1}(2-x)\right]_1^2 = \frac{\pi}{2}-0 = \frac{\pi}{2}$

9 a $\dfrac{6x^2}{4+x^6}$ b $\frac{1}{6}\tan^{-1}\frac{x^3}{2}+C$

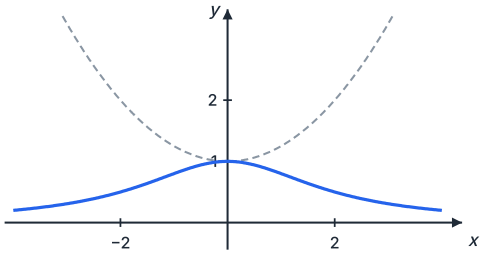
10 a 0 b 0 c $\frac{3\pi}{4}$ d 0 e 0 f 18π (half the area of a circle of radius 6)

11 a (i) 0 b (i) $f(0)=0$ and $f'(x)<0$ for $x>0$, so f is decreasing from 0. (ii) $\dfrac{\pi-2}{8}$

12 **c** $\frac{\pi}{3}$ unit² **d** π unit²



13 **a** the y-axis (it is an even function) **b** domain all real x, range $0 < y \leq 1$ **d** 0 **e** π unit² **f** $4\tan^{-1}\frac{a}{2}$ unit² **g** 2π unit²



14 **b** The formula gives about 0.153 unit². **c** The integrand lies between $\frac{1}{9}$ and $\frac{1}{4}$ on $[0, 7]$, so the area lies between $\frac{7}{9}$ and $\frac{7}{4}$, much larger than 0.153. The "primitive" is undefined at $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ inside the interval, which makes the calculation invalid.

Chapter 12 Further Calculus - Homework Answers

Lesson 12B | Inverse trigonometric functions: integrating

Past-Paper Homework - Worked Solutions

PP 1 **Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q8**

Let $u = 2f(x)$, so $u' = 2f'(x)$. The integrand is $\dfrac{u'}{1+u^2}$, whose primitive is $\tan^{-1} u = \tan^{-1}\left(2f(x)\right)$. **Answer (D).**

PP 2 **Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(d)**

$$\int \frac{1}{9+16x^2}\,dx = \frac{1}{3\times 4}\tan^{-1}\frac{4x}{3} + C = \tfrac{1}{12}\tan^{-1}\frac{4x}{3} + C$$

PP 3 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(a)**

$$\int \frac{1}{\sqrt{4-9x^2}}\,dx = \tfrac{1}{3}\sin^{-1}\frac{3x}{2} + C$$

PP 4 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(a)(ii)**

$$\int_0^{\frac{1}{3}} \frac{-1}{\sqrt{\frac{1}{9}-x^2}}\,dx = \Big[\cos^{-1} 3x\Big]_0^{\frac{1}{3}} = \cos^{-1}1 - \cos^{-1}0 = 0 - \tfrac{\pi}{2} = -\tfrac{\pi}{2}$$

(Equivalently $-\Big[\sin^{-1}3x\Big]_0^{\frac{1}{3}} = -\frac{\pi}{2}$.)

PP 5 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(f)**

$$f(x) = \int \frac{-3}{\sqrt{16-9x^2}}\,dx = 3\times\tfrac{1}{3}\cos^{-1}\frac{3x}{4} + C = \cos^{-1}\frac{3x}{4} + C$$

At $\left(0, -\frac{\pi}{2}\right)$: $\frac{\pi}{2} + C = -\frac{\pi}{2}$, so $C = -\pi$ and $f(x) = \cos^{-1}\dfrac{3x}{4} - \pi$. (Equivalently $f(x) = -\sin^{-1}\dfrac{3x}{4} - \dfrac{\pi}{2}$.)

PP 6 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(d)**

(i)

$$\frac{d}{dx}\left(x\sin^{-1}x + \sqrt{1-x^2}\right) = \sin^{-1}x + \frac{x}{\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}} = \sin^{-1}x$$

(ii)

$$\int_0^1 2\sin^{-1}x\,dx = 2\Big[x\sin^{-1}x + \sqrt{1-x^2}\Big]_0^1 = 2\left(\tfrac{\pi}{2} + 0 - 0 - 1\right) = \pi - 2$$

PP 7 **Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)**

(i) Quotient rule for the first term:

$$\begin{aligned}\frac{d}{dx}\left(\frac{2x}{4+x^2}\right) + \frac{d}{dx}\tan^{-1}\frac{x}{2} &= \frac{2(4+x^2) - 2x\cdot 2x}{(4+x^2)^2} + \frac{2}{4+x^2} \\ &= \frac{8-2x^2}{(4+x^2)^2} + \frac{2(4+x^2)}{(4+x^2)^2} = \frac{16}{(4+x^2)^2}\end{aligned}$$

(ii)

$$\int_0^{2\sqrt{3}} \frac{dx}{(4+x^2)^2} = \tfrac{1}{16}\left[\frac{2x}{4+x^2} + \tan^{-1}\frac{x}{2}\right]_0^{2\sqrt{3}} = \tfrac{1}{16}\left(\frac{4\sqrt{3}}{16} + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{64} + \frac{\pi}{48}$$

Chapter 12 Further Calculus - Homework Answers

Lesson 12C | Further trigonometric integrals

Lesson Examples

C2 **Tutor demo 1.** a $\left[\frac{1}{2}x+\frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{4}}=\frac{\pi}{8}+\frac{1}{4}$ b $\left[\frac{1}{2}x-\frac{1}{2}\sin x\right]_0^{\pi}=\frac{\pi}{2}$

Your turn 1. a $\frac{\pi}{4}$ b $\left[\frac{1}{2}x-\frac{1}{8}\sin 4x\right]_0^{\frac{\pi}{2}}=\frac{\pi}{4}$

Tutor demo 2. $\int(\sec^2 x-1)\,dx=\tan x-x+C$ **Your turn 2.** $1-\frac{\pi}{4}$

C3 **Tutor demo.** a $\int\frac{\sin x}{\cos x}\,dx=-\ln|\cos x|+C$ b $\frac{1}{4}\sin^4 x+C$

Your turn. a $\frac{1}{3}\ln|\sin 3x|+C$ b $-\frac{1}{6}\cos^6 x+C$

C4 **Tutor demo.** $\frac{1}{2}\int(\sin 8x+\sin 2x)\,dx=-\frac{1}{16}\cos 8x-\frac{1}{4}\cos 2x+C$

Your turn. $\int(\cos 4x+\cos 2x)\,dx=\frac{1}{4}\sin 4x+\frac{1}{2}\sin 2x+C$

Textbook Homework

2 a $\frac{2+\sqrt{3}}{4}$ b $\frac{2+\sqrt{3}}{4}$ c $-\frac{1}{4}$ d $\frac{2-\sqrt{2}}{4}$

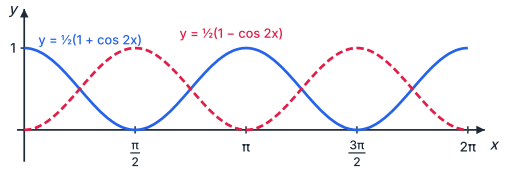
3 a $\frac{1}{2}x-\frac{1}{4}\sin 2x+C$ b $\frac{1}{2}x-\frac{1}{8}\sin 4x+C$ c $\frac{1}{2}x-\sin\frac{1}{2}x+C$ d $\frac{1}{2}x-\frac{1}{12}\sin 6x+C$

4 a $\frac{1}{2}x+\frac{1}{4}\sin 2x+C$ b $\frac{1}{2}x+\frac{1}{24}\sin 12x+C$ c $\frac{1}{2}x+\frac{1}{2}\sin x+C$ d $\frac{1}{2}x+\frac{1}{40}\sin 20x+C$

5 a $\frac{\pi}{2}$ b $\frac{1}{8}(\pi+2)$ c $\frac{1}{12}(\pi-3)$ d $\frac{1}{32}(\pi+2\sqrt{2})$ e $\frac{1}{24}(4\pi+9)$ f $\frac{1}{24}(2\pi-3\sqrt{3})$

6 a $-\frac{1}{10}\cos 5x-\frac{1}{2}\cos x+C$ b $\frac{1}{4}\cos 2x-\frac{1}{8}\cos 4x+C$ c $\frac{2\sqrt{2}}{3}$ d $\frac{\sqrt{3}}{28}$

7 b The two curves are reflections of each other in $y=\frac{1}{2}$, and their heights add to 1 at every x.



8 a $\frac{1}{4}\sin^4 x$ b $\frac{1}{7}\sin^7 x$ c $-\frac{1}{6}\cos^6 x$ d $-\frac{1}{9}\cos^9 x$ e $-\cos e^x$ f $\frac{1}{5}\sin 5e^x$ g $-\log_e|\cos x|$ h $\frac{1}{7}\log_e|\sin 7x|$ (each $+C$)

9 a $\cos x\sin x=\frac{1}{2}\sin 2x$, so $-\frac{1}{2}\le y\le\frac{1}{2}$. b (i) $-\frac{1}{4}\cos 2x+C$ (ii) $\frac{1}{2}\sin^2 x+D$ using $u=\sin x$, and $-\frac{1}{2}\cos^2 x+E$ using $u=\cos x$. c The constants C, D, E are different; the answers differ by constants, by the Pythagorean identity and the $\cos 2x$ formulae.

10 b $\cos^4 x=\frac{3}{8}+\frac{1}{2}\cos 2x+\frac{1}{8}\cos 4x$ c (i) $\frac{3\pi}{8}$ (ii) $\frac{1}{32}(3\pi+8)$ d $\frac{5}{24}$

11 a $\frac{1}{2}\tan 2x-x+C$ b $-2\cot\frac{1}{2}x-x+C$ c $\sqrt{3}-1-\frac{\pi}{12}$ d $\frac{1}{4}\sqrt{3}-\frac{\pi}{12}$

12 a $\frac{1}{2}\tan^2 x+C$ b $x-\sin x+C$ c $\tan x+\sin x+C$

13 b (i) $x=0,\pi,2\pi$ (ii) $0<x<\pi$ or $\pi<x<2\pi$ (iii) no values of x c because $-\frac{1}{4}\le\frac{1}{4}\sin 2x\le\frac{1}{4}$ d $\left(\frac{\pi}{2},\frac{\pi}{4}\right)$ and $\left(\frac{3\pi}{2},\frac{3\pi}{4}\right)$ are points of inflection, while $(0,0)$, $\left(\pi,\frac{\pi}{2}\right)$ and $(2\pi,\pi)$ are stationary (horizontal) points of inflection. f (i) $k=3\pi$ (ii) $k=n\pi$

Chapter 12 Further Calculus - Homework Answers

Lesson 12C | Further trigonometric integrals

Past-Paper Homework - Worked Solutions

PP 1 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q1

$$\int \sin^2 2x\,dx = \int \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 4x\right)dx = \frac{x}{2} - \frac{1}{8}\sin 4x + C$$

Answer A. C forgets to divide by 4 when integrating $\cos 4x$; B and D have the wrong sign.

PP 2 North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q3

$2\sin 3x\cos 4x = \sin 7x + \sin(-x) = \sin 7x - \sin x$. **Answer A.**

PP 3 Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q13(b)

$$\int \sin^2 3x\,dx = \int \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 6x\right)dx = \frac{x}{2} - \frac{1}{12}\sin 6x + C$$

PP 4 Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q14(a)

$$\sin^2 x\cos^2 x = \left(\tfrac{1}{2}\sin 2x\right)^2 = \tfrac{1}{4}\sin^2 2x = \tfrac{1}{8}(1-\cos 4x)$$

$$\int \sin^2 x\cos^2 x\,dx = \frac{x}{8} - \frac{1}{32}\sin 4x + C$$

PP 5 Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(e)

$$\cos^2 2x = \tfrac{1}{2} + \tfrac{1}{2}\cos 4x$$

$$\int_0^{\frac{\pi}{6}} \cos^2 2x\,dx = \left[\tfrac{1}{2}x + \tfrac{1}{8}\sin 4x\right]_0^{\frac{\pi}{6}} = \frac{\pi}{12} + \frac{1}{8}\times\frac{\sqrt{3}}{2} = \frac{\pi}{12} + \frac{\sqrt{3}}{16}$$

PP 6 Cherrybrook Technology High School, 2020, Year 12 AP4 Mathematics Extension 1 (Trial), Q13(a)

$$\int_0^{2\pi} \sin^2\frac{x}{2}\,dx = \int_0^{2\pi} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos x\right)dx = \left[\tfrac{1}{2}x - \tfrac{1}{2}\sin x\right]_0^{2\pi} = \pi$$

PP 7 Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(d)

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2 x\,dx = \left[\tfrac{1}{2}x + \tfrac{1}{4}\sin 2x\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \left(\tfrac{\pi}{6} + \tfrac{1}{4}\cdot\tfrac{\sqrt{3}}{2}\right) - \left(\tfrac{\pi}{12} + \tfrac{1}{4}\cdot\tfrac{\sqrt{3}}{2}\right) = \frac{\pi}{12}$$

PP 8 The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(a)

Reverse chain rule with $u = \sin x$, $u' = \cos x$:

$$\int_0^{\frac{\pi}{2}} \sin^2 x\cos x\,dx = \left[\tfrac{1}{3}\sin^3 x\right]_0^{\frac{\pi}{2}} = \tfrac{1}{3}$$

PP 9 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(a)

$$\int \sin 8x\cos 5x\,dx = \tfrac{1}{2}\int (\sin 13x + \sin 3x)\,dx = -\frac{1}{26}\cos 13x - \frac{1}{6}\cos 3x + C$$

Chapter 12 Further Calculus - Homework Answers

Lesson 12D | Integration by substitution

Lesson Examples

C1 **Tutor demo.** $du = 3x^2\,dx$: $\frac{1}{3}\int u^4\,du = \frac{1}{15}(1+x^3)^5 + C$ **Your turn.** $du = 2x\,dx$: $\frac{1}{2}\int u^5\,du = \frac{1}{12}(x^2+3)^6 + C$

C2 **Tutor demo 1.** $du = \cos x\,dx$: $\int u^{\frac{1}{2}}\,du = \frac{2}{3}(1+\sin x)^{\frac{3}{2}} + C$
Tutor demo 2. $x = u-2$, $dx = du$: $\int (u-2)u^{\frac{1}{2}}\,du = \frac{2}{5}(x+2)^{\frac{5}{2}} - \frac{4}{3}(x+2)^{\frac{3}{2}} + C$
Your turn. $x = u+1$: $\int (u+1)u^{\frac{1}{2}}\,du = \frac{2}{5}(x-1)^{\frac{5}{2}} + \frac{2}{3}(x-1)^{\frac{3}{2}} + C$

C3 **Tutor demo.** u from 1 to 2: $\frac{1}{3}\int_1^2 u^{-2}\,du = \frac{1}{3}\Big[-u^{-1}\Big]_1^2 = \frac{1}{6}$ **Your turn.** u from 1 to $\frac{1}{2}$:

$$-\int_1^{\frac{1}{2}} u^3\,du = \int_{\frac{1}{2}}^1 u^3\,du = \frac{15}{64}$$

Textbook Homework

1 **c** $\frac{1}{4}(1+x^2)^4 + C$

2 **a** $\frac{1}{4}(2x+3)^4 + C$ **b** $\dfrac{-1}{1+x^2} + C$ **c** $2\sqrt{3x-5} + C$ **d** $\log_e(1+x^4) + C$

3 **c** $-\sqrt{1-x^2} + C$

4 **a** $\frac{1}{24}(x^4+1)^6 + C$ **b** $\frac{2}{9}(x^3-1)^{\frac{3}{2}} + C$ **c** $\dfrac{-1}{(1+\sqrt{x})^2} + C$ **d** $\frac{1}{6}\tan^3 2x + C$

5 **a** $\frac{65}{12}$ **b** $\frac{1}{3}$ **c** $\frac{1}{24}$ **d** 2 **e** $\frac{1}{10}$ **f** $\frac{\pi^4}{64}$

6 $\frac{\pi}{12}$ units^2

7 **a** $\log_e \frac{3}{2}$ **b** $\frac{\pi}{4}$ **c** $\frac{2}{3}$

8 **a** $\sqrt{1+e^{2x}} + C$ **b** $\ln(\ln x) + C$

9 $y = \frac{1}{2}\tan^{-1} e^{2x}$

10 **b** (i) $\dfrac{2}{\ln 2}$ (ii) $\frac{1}{5}\left(4\sqrt{2}-1\right)$

11 $\ln 2$

12 **a** $2\sin^{-1}\sqrt{x} + C_1$ **b** $\sin^{-1}(2x-1) + C_2$ **c** The two primitives differ by a constant; put $x = \frac{1}{2}$ to find it: $2 \times \frac{\pi}{4} - 0 = \frac{\pi}{2}$.

13 With $u = x - \frac{1}{x}$: $du = \left(1 + \frac{1}{x^2}\right)dx$, and dividing top and bottom by x^2, the integrand becomes $\dfrac{1}{u^2+2}$. The limits become $u = 0$ and $u = \sqrt{2}$, giving $\frac{1}{\sqrt{2}}\tan^{-1} 1 = \dfrac{\pi}{4\sqrt{2}}$.

Chapter 12 Further Calculus - Homework Answers

Lesson 12D | Integration by substitution

Past-Paper Homework - Worked Solutions

PP 1 North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q2

$u = 1 + 3x$ gives $x = \frac{1}{3}(u-1)$ and $dx = \frac{1}{3}\,du$, so
$\int x\sqrt{1+3x}\,dx = \int \frac{1}{3}(u-1)\sqrt{u} \times \frac{1}{3}\,du = \frac{1}{9}\int (u-1)\sqrt{u}\,du$. **Answer A.**

PP 2 Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q8

$du = -\sin x\,dx$; when $x = 0$, $u = 1$; when $x = \frac{\pi}{3}$, $u = \frac{1}{2}$. So the integral is $-\int_1^{\frac{1}{2}} u^5\,du = \int_{\frac{1}{2}}^1 u^5\,du$. **Answer (D).** (B) has the minus sign and the swapped limits together; (A) and (C) keep the x-limits.

PP 3 Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(a)

$u = x^2 - 1,\ du = 2x\,dx$: $\int 2x\cos(x^2-1)\,dx = \int \cos u\,du = \sin(x^2-1) + C$

PP 4 James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(a)

$u = \sqrt{x},\ du = \dfrac{dx}{2\sqrt{x}}$: $\int \dfrac{\sin\sqrt{x}}{\sqrt{x}}\,dx = \int 2\sin u\,du = -2\cos\sqrt{x} + C$

PP 5 North Sydney Boys High School, 2023, Year 12 Mathematics Extension 1 Assessment Task 3, Q11(c)

$u = \tan^{-1}x,\ du = \dfrac{dx}{1+x^2}$: $\int \dfrac{e^{\tan^{-1}x}}{1+x^2}\,dx = \int e^u\,du = e^{\tan^{-1}x} + C$

PP 6 Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(e)

$u = x^3$, $du = 3x^2\,dx$; when $x = 0$, $u = 0$; when $x = 1$, $u = 1$.
$\int_0^1 x^2 e^{x^3}\,dx = \frac{1}{3}\int_0^1 e^u\,du = \frac{1}{3}\Big[e^u\Big]_0^1 = \frac{1}{3}(e-1)$

PP 7 Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q12(a)

$u = \tan x$, $du = \sec^2 x\,dx$; when $x = \frac{\pi}{4}$, $u = 1$; when $x = \frac{\pi}{3}$, $u = \sqrt{3}$.
$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x\sec^2 x\,dx = \int_1^{\sqrt{3}} u^2\,du = \Big[\frac{1}{3}u^3\Big]_1^{\sqrt{3}} = \sqrt{3} - \frac{1}{3}$

PP 8 Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q12(a)

$u = 1 - x$, $du = -dx$, $x = 1 - u$; when $x = 0$, $u = 1$; when $x = 1$, $u = 0$.
$\int_0^1 x(1-x)^7\,dx = -\int_1^0 (1-u)u^7\,du = \int_0^1 \big(u^7 - u^8\big)du = \frac{1}{8} - \frac{1}{9} = \frac{1}{72}$

PP 9 St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(f)

$u = 2x^3 + 3x$, $du = 3(2x^2+1)\,dx$; when $x = 0$, $u = 0$; when $x = 1$, $u = 5$.
$\int_0^1 \dfrac{2x^2+1}{25 + (2x^3+3x)^2}\,dx = \frac{1}{3}\int_0^5 \dfrac{du}{25+u^2} = \frac{1}{3} \times \frac{1}{5}\Big[\tan^{-1}\frac{u}{5}\Big]_0^5 = \frac{1}{15} \times \frac{\pi}{4} = \dfrac{\pi}{60}$

Chapter 12 Further Calculus - Homework Answers

Lesson 12E | Further integration by substitution

Lesson Examples

C1 **Tutor demo.** $dx = 2u\,du$, $\sqrt{x+1} = u$, u from 1 to 2: $\displaystyle\int_1^2 \frac{u^2-1}{u}\,2u\,du = 2\left[\tfrac{1}{3}u^3 - u\right]_1^2 = \tfrac{8}{3}$

Your turn. u from 1 to 3: $2\left[\tfrac{1}{3}u^3 - u\right]_1^3 = 2\left(6 + \tfrac{2}{3}\right) = \tfrac{40}{3}$

C2 **Tutor demo.** $dx = 3\cos\theta\,d\theta$, θ from 0 to $\frac{\pi}{6}$: $\displaystyle\int_0^{\frac{\pi}{6}} 9\cos^2\theta\,d\theta = 9\left[\tfrac{1}{2}\theta + \tfrac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{6}} = \tfrac{3\pi}{4} + \tfrac{9\sqrt{3}}{8}$

Your turn. θ from 0 to $\frac{\pi}{4}$: $4\left[\tfrac{1}{2}\theta + \tfrac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{4}} = \tfrac{\pi}{2} + 1$

Textbook Homework

1 **c** $\frac{1}{7}(x-1)^7 + \frac{1}{6}(x-1)^6 + C$

2 **a** $\frac{2}{3}(x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C$ **b** $\ln|x-1| - \dfrac{1}{x-1} + C$

3 **c** $\frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + C$

4 **a** $\frac{2}{7}(x+1)^{\frac{7}{2}} - \frac{4}{5}(x+1)^{\frac{5}{2}} + \frac{2}{3}(x+1)^{\frac{3}{2}} + C$ **b** $\frac{4}{3}(x+1)^{\frac{3}{2}} + 2(x+1)^{\frac{1}{2}} + C$

5 **a** $(x+2) - 4\ln(x+2) + C$ **b** $\frac{1}{3}(2x-1)^{\frac{3}{2}} + 2(2x-1)^{\frac{1}{2}} + C$ **c** $2(1+\sqrt{x}) - 2\ln(1+\sqrt{x}) + C$

6 **a** $\frac{49}{20}$ **b** $\frac{8}{9}$ **c** $\frac{128}{15}$ **d** $4 - 6\ln\frac{5}{3}$

7 **a** $\sin^{-1}\dfrac{x+2}{3} + C$ **b** (i) $\dfrac{1}{\sqrt{3}}\tan^{-1}\dfrac{x+1}{\sqrt{3}} + C$ (ii) $\frac{\pi}{6}$

8 **b** (i) $\frac{1}{3}\tan^{-1}\frac{x}{3} + C$ (ii) $\frac{1}{2}\sin^{-1}2x + C$ (iii) $\frac{\pi}{6}$

9 **b** (i) $\dfrac{x}{4\sqrt{4+x^2}} + C$ (ii) π

10 $y = \sqrt{x^2-9} - 3\tan^{-1}\dfrac{\sqrt{x^2-9}}{3}$

11 $\frac{1}{3}\left(6\sqrt{3} - 7\sqrt{2}\right)$ units²

12 $\displaystyle 2\int_{-r}^{r}\sqrt{r^2-x^2}\,dx = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} r^2\cos^2\theta\,d\theta = r^2\left[\theta + \tfrac{1}{2}\sin 2\theta\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \pi r^2$

13 **b** Adding the two equal forms gives $\displaystyle 2I = \int_{-2}^{2} x^2\,dx = \tfrac{16}{3}$, so $I = \frac{8}{3}$.

Chapter 12 Further Calculus - Homework Answers

Lesson 12E | Further integration by substitution

Past-Paper Homework - Worked Solutions

PP 1 The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(b)

$x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$, $\sqrt{4-x^2} = 2\cos\theta$. When $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \frac{\pi}{6}$.

$$\int_0^1 \sqrt{4-x^2}\,dx = \int_0^{\frac{\pi}{6}} 4\cos^2\theta\,d\theta = \left[2\theta + \sin 2\theta\right]_0^{\frac{\pi}{6}} = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

PP 2 Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d)

$x = u^2 - 1$, $dx = 2u\,du$, $x + 2 = u^2 + 1$. When $x = 0$, $u = 1$; when $x = 3$, $u = 2$.

$$\int_0^3 \frac{x+2}{\sqrt{x+1}}\,dx = \int_1^2 \frac{u^2+1}{u}\,2u\,du = 2\left[\tfrac{1}{3}u^3 + u\right]_1^2 = 2\left(\tfrac{14}{3} - \tfrac{4}{3}\right) = \frac{20}{3}$$

PP 3 Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d)

$dx = 2u\,du$, $\sqrt{x+1} = u$; u from 1 to 2.

$$\int_0^3 x\sqrt{x+1}\,dx = \int_1^2 (u^2-1)u \cdot 2u\,du = 2\left[\tfrac{1}{5}u^5 - \tfrac{1}{3}u^3\right]_1^2 = 2\left(\tfrac{31}{5} - \tfrac{7}{3}\right) = \frac{116}{15}$$

PP 4 Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q12(c)

$x = \frac{1}{4}(u^2+5)$, so $dx = \frac{1}{2}u\,du$ and $\sqrt{4x-5} = u$.

$$\begin{aligned}\int 3x\sqrt{4x-5}\,dx &= \int 3\cdot\tfrac{1}{4}(u^2+5)\cdot u\cdot\tfrac{1}{2}u\,du = \tfrac{3}{8}\int \left(u^4+5u^2\right)du \\ &= \tfrac{3}{40}u^5 + \tfrac{5}{8}u^3 + C = \tfrac{3}{40}(4x-5)^{\frac{5}{2}} + \tfrac{5}{8}(4x-5)^{\frac{3}{2}} + C\end{aligned}$$

PP 5 Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q13(d)

$x = \sin\theta$, $dx = \cos\theta\,d\theta$, $\sqrt{1-x^2} = \cos\theta$; θ from 0 to $\frac{\pi}{6}$.

$$\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx = \int_0^{\frac{\pi}{6}} \sin^2\theta\,d\theta = \left[\tfrac{1}{2}\theta - \tfrac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{6}} = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$$

PP 6 Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q11(d)

(i) $1 - \dfrac{1}{1+u^2} = \dfrac{1+u^2-1}{1+u^2} = \dfrac{u^2}{1+u^2}$.

(ii) $x = u^2$, $dx = 2u\,du$, $\sqrt{x} = u$:

$$\int \frac{\sqrt{x}}{1+x}\,dx = \int \frac{u}{1+u^2}\,2u\,du = 2\int\left(1 - \frac{1}{1+u^2}\right)du = 2u - 2\tan^{-1}u + C = 2\sqrt{x} - 2\tan^{-1}\sqrt{x} + C$$

Chapter 12 Further Calculus - Homework Answers

Lesson 12F | Volumes of rotation

Lesson Examples

C1 **Tutor demo.** $\pi \int_1^3 (9 - x^2)\,dx = \pi\left[9x - \frac{1}{3}x^3\right]_1^3 = \frac{28\pi}{3}$ u³ **Your turn.** $\pi \int_0^4 x\,dx = 8\pi$ u³

C2 **Tutor demo.** $x^2 = y$: $\pi \int_1^4 y\,dy = \frac{15\pi}{2}$ u³ **Your turn.** $x^2 = y^4$: $\pi \int_0^2 y^4\,dy = \frac{32\pi}{5}$ u³

C3 **Tutor demo.** The curves meet at $x = 0$ and 1, with $x > x^2$ between: $\pi \int_0^1 (x^2 - x^4)dx = \frac{2\pi}{15}$ u³

Your turn. In terms of y, the line is $x = y$ and the parabola is $x = \sqrt{y}$, with $\sqrt{y} > y$ for $0 < y < 1$:

$\pi \int_0^1 (y - y^2)dy = \frac{\pi}{6}$ u³

Textbook Homework

1 **b** radius 9, height 3, $V = 81\pi$ u³

2 **b** radius 3, $V = 36\pi$ u³

3 **a** 16π **b** $\frac{32\pi}{5}$ **c** 6π **d** $\frac{16\pi}{3}$ **e** 9π (all u³)

4 **a** $\frac{256\pi}{3}$ **b** $\frac{3093\pi}{5}$ **c** $\frac{256\pi}{3}$ **d** $\frac{243\pi}{5}$ (all u³)

5 $\frac{\pi}{2}(e^2 - 1)$ u³

6 $\pi \ln 2$ u³

7 $\pi \ln 6$ u³

8 **a** $\tan^2 x = \sec^2 x - 1$ **b** $\pi\left(\sqrt{3} - \frac{\pi}{3}\right)$ u³

9 **a** $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ **b** $\frac{\pi^2}{4}$ u³

10 **a** $\frac{296\pi}{3}$ **b** $\frac{119\pi}{6}$ **c** $\frac{16\pi}{105}$ (all u³)

11 **a** $\frac{\pi}{3}$ **b** $\frac{28\pi}{15}$ **c** $\frac{\pi}{2}$ (all u³)

12 **a** 256π u³ **b** 128π u³ **c** 128π u³

13 **a** $\pi \int_0^h \frac{r^2x^2}{h^2}\,dx = \frac{1}{3}\pi r^2 h$ **b** $\pi \int_0^h r^2\,dx = \pi r^2 h$ **c** (i) $\pi \int_{-r}^{r} (r^2 - x^2)\,dx = \frac{4}{3}\pi r^3$ (ii)
$\pi \int_{r-h}^{r} (r^2 - x^2)\,dx = \frac{1}{3}\pi h^2(3r - h)$

14 **b** (i) $\frac{2\pi}{35}$ u³ (ii) $\frac{\pi}{10}$ u³

15 **a** They meet at $(1, 5)$ and $(5, 1)$. **b** $\frac{64\pi}{3}$ u³

16 **a** $(0, 0)$ and $(1, 1)$ **c** the cone formed by rotating the line $y = x$ from $x = 0$ to $x = 1$ about the x-axis **d** $\frac{\pi}{3}$ u³

Chapter 12 Further Calculus - Homework Answers

Lesson 12F | Volumes of rotation

Past-Paper Homework - Worked Solutions

PP 1 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q5**

$V = \pi \int_0^{\frac{\pi}{3}} \sin^2 x \, dx = \dfrac{\pi}{2} \int_0^{\frac{\pi}{3}} (1 - \cos 2x) \, dx$. **Answer B.** A and D have the sign reversed (they would give a negative volume); C uses $\cos^2 x$.

PP 2 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q2**

The curve is the upper half of the circle $(x+1)^2 + y^2 = 16$, meeting the x-axis at $x + 1 = \pm 4$, that is, $x = -5$ and $x = 3$. Then $V = \pi \int_{-5}^{3} y^2 \, dx = \pi \int_{-5}^{3} \left(16 - (x+1)^2\right) dx$. **Answer D.**

PP 3 **Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(a)**

$$V = \pi \int_2^8 \frac{1}{x} \, dx = \pi \Big[\ln x \Big]_2^8 = \pi \ln 4 = 2\pi \ln 2 \text{ u}^3$$

PP 4 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(b)**

$y = \ln x$ gives $x = e^y$, so $x^2 = e^{2y}$, for $0 \le y \le \ln 4$.

$$V = \pi \int_0^{\ln 4} e^{2y} \, dy = \frac{\pi}{2} \Big[e^{2y} \Big]_0^{\ln 4} = \frac{\pi}{2}(16 - 1) = \frac{15\pi}{2} \text{ u}^3$$

PP 5 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(d)**

Intersections: $(x-1)^2 = x + 1$ gives $x^2 - 3x = 0$, so $x = 0$ or $x = 3$. The line is above the parabola for $0 < x < 3$, and both are non-negative there.

$$V = \pi \int_0^3 \left((x+1)^2 - (x-1)^4 \right) dx = \pi \Big[\tfrac{1}{3}(x+1)^3 - \tfrac{1}{5}(x-1)^5 \Big]_0^3$$

$$= \pi \left[\left(\tfrac{64}{3} - \tfrac{32}{5} \right) - \left(\tfrac{1}{3} + \tfrac{1}{5} \right) \right] = \pi \left(21 - \tfrac{33}{5} \right) = \frac{72\pi}{5} \text{ u}^3$$

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)**

$$V = \pi \int_0^1 \frac{4}{x^2 + 3} \, dx = 4\pi \times \frac{1}{\sqrt{3}} \Big[\tan^{-1} \frac{x}{\sqrt{3}} \Big]_0^1 = \frac{4\pi}{\sqrt{3}} \times \frac{\pi}{6} = \frac{2\pi^2}{3\sqrt{3}} = \frac{2\sqrt{3}\,\pi^2}{9} \text{ u}^3$$

PP 7 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q11(c)**

(i) Domain $-1 \le x \le 1$, range $-\pi \le y \le \pi$; an increasing curve through the origin with endpoints $(-1, -\pi)$ and $(1, \pi)$, and vertical tangents at the endpoints.

(ii) $y = 2\sin^{-1} x$ gives $x = \sin \frac{y}{2}$, for $0 \le y \le \frac{\pi}{2}$.

$$V = \pi \int_0^{\frac{\pi}{2}} \sin^2 \tfrac{y}{2} \, dy = \pi \int_0^{\frac{\pi}{2}} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos y \right) dy = \pi \Big[\tfrac{1}{2}y - \tfrac{1}{2}\sin y \Big]_0^{\frac{\pi}{2}} = \pi \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi^2}{4} - \frac{\pi}{2} \text{ u}^3$$

Chapter 12 Further Calculus - Homework Answers

Chapter 12 Review | Chapter review exercise

1 **a** $\dfrac{3}{\sqrt{1-9x^2}}$ **b** $\dfrac{3}{9+x^2}$ **c** $\dfrac{1}{\sqrt{2x-x^2}}$ **d** $2x\tan^{-1}x+\dfrac{x^2}{1+x^2}$ **e** $\dfrac{2}{x^2+4x+8}$ **f** $\dfrac{-1}{\sqrt{x^4-x^2}}$

3 **a** Each has derivative $\dfrac{-1}{\sqrt{1-x^2}}$. **b** The functions differ by at most a constant. The constant is zero, so $\cos^{-1}x=\sin^{-1}\sqrt{1-x^2}$ for $0\le x\le 1$.

4 **a** $\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3}}+C$ **b** $\sin^{-1}\frac{x}{\sqrt{3}}+C$ **c** $\frac{1}{6}\tan^{-1}\frac{2x}{3}+C$ **d** $\frac{1}{3}\cos^{-1}\frac{3x}{4}+C$

5 **a** $\frac{\pi}{36}$ **b** $\frac{\pi}{3}$

6 **a** $\frac{1}{2}x+\frac{1}{4}\sin 2x+C$ **b** $\frac{1}{2}x-\frac{1}{4}\sin 2x+C$ **c** $\frac{1}{2}x+\frac{1}{8}\sin 4x+C$ **d** $\frac{1}{2}x-\frac{1}{16}\sin 8x+C$

7 **a** $\frac{\pi}{6}$ **b** $\frac{1}{12}(\pi+3)$

9 **a** $\frac{1}{6}(5x-1)^6+C$ **b** $\frac{1}{3}(x^2+2)^3+C$ **c** $\dfrac{-1}{x^4+1}+C$ **d** $\frac{1}{2}\sqrt{4x+3}+C$ **e** $\frac{1}{3}\sin^3x+C$ **f** $\frac{1}{4}\tan^4x+C$

10 **a** $\frac{1}{15}$ **b** $\frac{1}{4}$ **c** $\frac{1}{3}$ **d** $\frac{1}{3}$ **e** e^2-e **f** $\frac{1}{2}\ln 2$

11 **a** $x-1+\ln(x-1)+C$ **b** $\frac{2}{3}(x+2)^{\frac{3}{2}}-6(x+2)^{\frac{1}{2}}+C$ **c** $\frac{1}{10}(2x+1)^{\frac{5}{2}}-\frac{1}{6}(2x+1)^{\frac{3}{2}}+C$ **d** $2(4+\sqrt{x})-8\ln(4+\sqrt{x})+C$

12 **a** $\frac{11}{30}$ **b** $4-3\ln 2$ **c** 36 **d** $\frac{13}{15}$

13 **a** $x\le 9, y\ge 0$ **c** 18 u² **d** (i) $\frac{81\pi}{2}$ u³ (ii) $\frac{648\pi}{5}$ u³

14 $4\pi\ln 2$ u³

15 $\frac{2688\pi}{5}$ u³

16 **a** $\cos^2 2x=\frac{1}{2}+\frac{1}{2}\cos 4x$

17 ≈ 8.49 u³

18 **b** $\pi(1-\ln 2)$ u³

19 **a** minimum turning point at $(1,2)$, maximum turning point at $(-1,-2)$ **c** $\frac{9\pi}{4}$ u³

20 **b** $72-\frac{9\pi}{2}$ u² **c** $\frac{5004\pi}{5}$ u³

21 **a** $\pi\displaystyle\int_1^3 2^{2x+2}\,dx$ **b** 180π u³ **c** $\dfrac{120\pi}{\ln 2}\approx 173\pi$ u³. The curve is concave up, so the trapezoidal rule overestimates.